


**Course: Transversal ISI Project (ING4187)**

# Project 2: Kingdom Communication Solutions (KCS) Media Studio - Web-Based AI Video Transcription and Multilingual Caption Editing

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# Contents

**List of Abbreviations and Glossary of Terms** **ii**

**2 Chapter 2: State of the Art (Observation Phase)** **1**
2.1 Introduction and Review Scope . . . 1
2.2 Review Questions, Search Method, and Mapping . . . 2
2.2.1 Literature Search Execution and Protocol . . . 2
2.2.2 Selection Criteria and PRISMA Flow . . . 2
2.3 Speech Recognition Foundation Models and Tonal Pitch Modeling . . . 3
2.3.1 The Commercial Technological Gap in Central Africa . . . 3
2.3.2 Architectural Causes of Model Failure on Cameroonian Speech . . . 4
2.3.3 The Character-Level CTC Solution . . . 5
2.4 The General Alphabet of Cameroonian Languages and Font Rendering . . . 5
2.4.1 The 1979 GACL Unified Writing System . . . 5
2.4.2 Technical Audit of Commercial Video Editors and WebVTT Renderers . . . 5
2.5 African NLP and Regional Village Dialect Continua . . . 6
2.5.1 The Reality of Village Dialect Continua . . . 6
2.5.2 Decoupled Dialect Architecture in KCS Media Studio . . . 6
2.6 Field Data Acquisition, 9-Parameter Nomenclature, and Data Governance . . . 6
2.6.1 Offline Mobile Collection with KoboToolbox and Mobile Tooling . . . 7
2.6.2 The 9-Parameter File Nomenclature and Speaker Leakage Prevention . . . 7
2.6.3 Informed Consent and Double-Validation Workflow . . . 7
2.7 Video Subtitle Editors, Bantu Text Expansion, and Browser Audio Waveforms . . . 7
2.7.1 Timeline Desynchronization from Bantu Text Expansion . . . 7
2.7.2 The Manual Subtitling Productivity Bottleneck . . . 8
2.7.3 Browser Waveforms and Automated Boundary Snapping . . . 8
2.8 Measuring System Success and Usability . . . 8
2.8.1 Objective Speech Recognition Scores . . . 8
2.8.2 Productivity and Usability Metrics . . . 8
2.9 Concrete Systems and Tooling Comparison Matrix . . . 8
2.10 Systematic Comparative Synthesis of Existing Solutions . . . 8
2.11 Evidence-Based Research Gap and 10 Practical System Challenges . . . 9
2.11.1 The Evidence-Based Research Gap . . . 9
2.11.2 10 Practical System Challenges for Chapter 3 . . . 9
2.11.3 Traceability Matrix: Connecting Chapter 1 to Chapter 3 . . . 11
2.12 Chapter Summary . . . 12

**Appendix: Systematic Literature Review Appraisal Table** **14**

**References** **15**

# List of Abbreviations and Glossary of Terms

Table 1: Glossary of Core Computational, Linguistic, and Methodological Terms

| Term / Acronym | Explanation and Practical Role in the Project |
|---|---|
| **ASR** | **Automatic Speech Recognition**: Computational systems that convert spoken acoustic signals into written text [1, 2]. |
| **CER / WER** | **Character / Word Error Rate**: Standard evaluation metrics for transcription accuracy. WER scores whole-word correctness, whereas CER measures character-level precision, which is indispensable for tonal languages where a single diacritic alters lexical meaning completely. |
| **CTC** | **Connectionist Temporal Classification**: An end-to-end neural sequence alignment algorithm mapping continuous audio frames directly to output character sequences without requiring external phonetic aligners [3, 4]. |
| **GACL / AGCL** | **General Alphabet of Cameroonian Languages**: The national standard orthography established in 1979 by Tadadjeu and Sadembouo [5] to transcribe all Cameroonian languages consistently using unified phonetic characters and tone diacritics. |
| **MMS / XLS-R** | **Massively Multilingual Speech / Cross-Lingual Speech Representations**: Self-supervised speech foundation models pre-trained across hundreds of languages and adaptable to local low-resource speech via downstream fine-tuning [6, 7]. |
| **NFC / NFD** | **Normalization Form C / D**: Unicode normalization standards. NFC combines base characters and diacritics into precomposed canonical code points, whereas NFD decomposes them into base glyphs followed by combining diacritical marks [8]. |
| **PRISMA** | **Preferred Reporting Items for Systematic Reviews and Meta-Analyses**: An evidence-based minimum guideline for transparent and reproducible systematic literature reviews [9]. |
| **Tone** | Contrastive vocal pitch register (high, mid, low) or contour. In Cameroonian tonal languages, pitch contrasts distinguish both lexical items and grammatical inflection. |
| **Diacritic Stacking** | Placement of one or more tone-marking diacritical accents directly above or below base phonetic characters, such as open-o (ɔ) or schwa (ə). |
| **Web Audio API** | High-performance browser-native audio processing API enabling direct client-side sound decoding, waveform rendering, and sub-15ms scrubbing [10]. |
| **WebVTT** | **Web Video Text Tracks**: W3C standard format (`.vtt`) for timed subtitles and video text track synchronization [11]. |
| **Nunamaker SDIS** | Systems Development in Information Systems research framework [12] progressing through observation, theory building, system design, and empirical evaluation. |

# 2 Chapter 2: State of the Art (Observation Phase)

## 2.1 Introduction and Review Scope

Chapter 1 established the foundational goals and context for the **Kingdom Communication Solutions (KCS) Media Studio**[1]. We documented two motivating real-world challenges: protecting our cultural oral heritage from digital disappearance (**MS1**), and ensuring that vital public information in health, education, and civic affairs reaches all citizens in languages they readily understand (**MS2**):

- **MS1 (Cultural Oral Heritage Preservation):** In Cameroon, cultural traditions and indigenous knowledge have historically been preserved through spoken storytelling [13]. When indigenous languages are absent from modern digital video formats, younger generations switch predominantly to official European languages, accelerating language loss across generations [14].
- **MS2 (Making Essential Information Understandable to Everyone):** During critical public communication events—such as healthcare campaigns against infectious diseases, agricultural extension advisories, and civic voter education—information broadcasts delivered solely in French or English fail to reach rural and peri-urban populations. Over 35% of rural citizens in Cameroon do not have fluent literacy in official European languages, yet they fully understand spoken local languages [15, 16]. Subtitled video offers an effective medium for knowledge transfer, yet creators currently lack assistive software, forcing exhausting manual transcription and alignment [17].

Chapter 1 also formulated our two core problem statements: standard speech AI models fail on Cameroonian tonal languages (**PS1**), and existing video editing software cannot display local letters or handle regional dialect differences (**PS2**). To address these issues, we set forth two guiding research questions: how to support regional village dialects without alienating local speakers (**RQ1**), and how to build a responsive browser-based video editor that integrates speech AI with local dictionaries (**RQ2**).

[1]The KCS Media Studio is a web-based platform engineered to provide AI speech-to-text transcription, dialect-adaptive caption editing, and synchronized video subtitling for indigenous Cameroonian languages.

This chapter carries out the **Observation phase** of the Nunamaker Systems Development in Information Systems (SDIS) framework [12]. It directly fulfills research objectives **RO1.1.O** and **RO2.1.O**:

- **RO1.1.O (Observation):** Interview native translators to document specific dialect variations, slang words, and spelling confusions in existing video tools.
- **RO2.1.O (Observation):** Benchmark client-side audio rendering latency and diagnose font-clipping bugs for GACL characters in WebVTT subtitle tracks.

According to Ethnologue (2021) [18], Cameroon hosts **281 living indigenous languages**. To construct a rigorous, representative empirical baseline, we focus our observation across three distinct linguistic groups:

1. **Ghɔmálá'**[2] is an Eastern Grassfields language spoken by more than 1.1 million people in the West Region, recognized for its complex system of contrastive pitch tones and expanded vowel inventory.
2. **Ewondo**[3] is a Beti-Pahuin Bantu language spoken by more than 2.5 million speakers across the Centre and South Regions and Cameroon's capital city.
3. **Basaa**[4] is a Narrow Bantu language of the A40 zone with rich oral literature, complex tonal patterns, and agglutinative noun-verb morphology.

This review examines foundation speech models, audits commercial subtitle editors, synthesizes comparative capabilities, and concludes with ten concrete engineering challenges ($C_1$–$C_{10}$) that directly guide the Chapter 3 system design.

[2]ISO 639-3 code: `bbj`. Spoken by over 1.1 million people in the West Region (Mifi, Koung-Khi, and Hauts-Plateaux divisions). Belongs to the Eastern Grassfields Bamileke branch of Southern Bantoid languages [19, 20].

[3]ISO 639-3 code: `ewo`; Guthrie code: A72. Spoken by over 2.5 million first- and second-language speakers across the Centre and South Regions, functioning as the historical vehicular language of Yaoundé [21, 22].

[4]ISO 639-3 code: `bas`; Guthrie code: A43. Spoken by roughly 800,000 people across the Littoral and Centre Regions, particularly in the Sanaga-Maritime and Nyong-et-Kéllé divisions [22, 23].

## 2.2 Review Questions, Search Method, and Mapping

To ensure our review examines every requirement identified in Chapter 1 thoroughly and objectively, we formulated **nine Review Questions** ($RQ_{Rev}$):

1. **$RQ_{Rev}1$ (Speech Foundation Models):** Which speech foundation models can be adapted to low-resource African languages, and how accurately do they transcribe local speech out of the box?
2. **$RQ_{Rev}2$ (Voice Pitch and Tonal Contrast):** Why do standard speech recognition models confuse words in Cameroonian tonal languages, and how does pitch alter lexical meaning in Ghɔmálá', Ewondo, and Basaa?
3. **$RQ_{Rev}3$ (Writing Systems and Diacritics):** Why does standard software fail when displaying stacked diacritics from the 1979 General Alphabet of Cameroonian Languages (GACL)?
4. **$RQ_{Rev}4$ (Regional Village Dialects):** How do vocabulary and pronunciation shift across village dialects, and how can software accommodate these variants without alienating native speakers?
5. **$RQ_{Rev}5$ (Field Data Acquisition):** How can teams collect pristine audio in rural areas lacking reliable cellular internet and electricity while securing documented informed consent?
6. **$RQ_{Rev}6$ (Audio Nomenclature and Data Governance):** How should field recordings be indexed to prevent train-test data leakage, and how does a two-tier validation workflow protect training data quality?
7. **$RQ_{Rev}7$ (Commercial Video Editors):** What specific functional gaps emerge when using suites like Adobe Premiere Pro, DaVinci Resolve, or YouTube Studio for Cameroonian languages?
8. **$RQ_{Rev}8$ (Bantu Text Expansion and Synchronization):** Why does translating dialogue into Bantu languages expand subtitle text by 30% to 50%, and how do client-side waveforms resolve timeline drift?
9. **$RQ_{Rev}9$ (Evaluation Metrics):** What objective accuracy scores and usability benchmarks are needed to verify that the platform genuinely improves translator productivity?

Table 2 maps these nine Review Questions directly to the Nunamaker Observation Objectives defined in Chapter 1.

### 2.2.1 Literature Search Execution and Protocol

On October 18, 2026, we performed a systematic search across five major electronic bibliographic databases: **IEEE Xplore**, **ACM Digital Library**, **ACL Anthology**, **Scopus**, and **Google Scholar**, focusing on literature published between 2006 and 2026. All search strings, timestamps, hit counts, and screening determinations were recorded in our open-access GitHub repository search log[5].

### 2.2.2 Selection Criteria and PRISMA Flow

We applied explicit inclusion and exclusion criteria to ensure scientific and technical relevance:

- **Inclusion Criteria:**
  - $IC_1$: Peer-reviewed conference or journal publications, authoritative international technical standards (e.g., W3C recommendations, Unicode Consortium specifications), and authoritative reference grammars or phonetic dictionaries.
  - $IC_2$: Explicit focus on low-resource speech recognition, African or Bantu tonal phonology, mobile language documentation, or web-based video captioning technology.
  - $IC_3$: Empirical evaluation, documented corpora, reproducible architectures, or validated usability metrics.
  - $IC_4$: Published or officially released between 2006 and 2026.
- **Exclusion Criteria:**
  - $EC_1$: Commercial vendor promotional whitepapers lacking peer review, verifiable methodology, or source access.
  - $EC_2$: Studies evaluating high-resource Indo-European languages exclusively without low-resource adaptation mechanisms.
  - $EC_3$: Extended abstracts lacking reproducible empirical results or technical depth.

[5]The complete machine-readable PRISMA search log is deposited at: `https://github.com/Abba-max/Media_Studio_Software_for_African_Languages`.

Table 2: Systematic Mapping of Review Questions to Nunamaker Observation Research Objectives

| Review Question | Thematic Focus Area | Chapter 1 Objective | Observation Deliverable and Scope |
|---|---|---|---|
| $RQ_{Rev}1$, $RQ_{Rev}2$ | Speech Foundation Models & Tone | **RO2.1.O** (Observation) | ASR diagnostic error baseline, pitch modeling gap, and CTC character decoding requirements. |
| $RQ_{Rev}3$, $RQ_{Rev}7$ | GACL Diacritics & Video Editors | **RO2.1.O** (Observation) | Subtitle editor typography audit, WebVTT rendering failures, and font-clipping diagnosis. |
| $RQ_{Rev}4$ | Regional Village Dialect Continua | **RO1.1.O** (Observation) | Sociolinguistic audit of lexical variation and translator usability barriers. |
| $RQ_{Rev}5$, $RQ_{Rev}6$ | Field Collection & Governance | **RO1.1.O** (Observation) | Offline mobile data collection, 9-parameter audio nomenclature, and train-test leakage control. |
| $RQ_{Rev}8$, $RQ_{Rev}9$ | Bantu Expansion & Usability | **RO2.1.O** (Observation) | Morphological expansion timing drift, browser Web Audio latency, and productivity benchmarking. |

  - $EC_4$: Research focusing strictly on physical television broadcast hardware or analog transmission protocols without relevance to digital web captioning.

Following the PRISMA framework [9], 342 initial records were identified, deduplicated, screened, and evaluated for full-text eligibility. Table 3 summarizes the search queries across the databases, and Table 4 details the PRISMA screening flow, yielding **41 included primary studies** as our empirical foundation.

## 2.3 Speech Recognition Foundation Models and Tonal Pitch Modeling

This section addresses $\mathbf{RQ_{Rev}1}$ and $\mathbf{RQ_{Rev}2}$ by examining modern speech foundation models and analyzing their failure modes on African tonal languages.

Over the past five years, automatic speech recognition has shifted toward self-supervised foundation architectures [2, 6, 7, 24]. Rather than training acoustic models from scratch, large neural networks are pre-trained on tens of thousands of hours of unlabelled speech audio. The network learns generalized acoustic representations of human speech, capturing phoneme boundaries, formant transitions, and coarticulation effects. Once pre-trained, the model can be fine-tuned on a target language using modest amounts of supervised audio.

### 2.3.1 The Commercial Technological Gap in Central Africa

Despite rapid global adoption, commercial cloud speech platforms overlook Cameroonian languages entirely [25]:

- **Commercial Cloud Speech APIs:** Official documentation for Google Cloud Speech-to-Text [26] and Microsoft Azure Cognitive Services Speech [27] confirms that neither vendor supports any of Cameroon's 281 indigenous languages.
- **OpenAI Whisper (Large-v3) [2]:** Whisper was trained on 680,000 hours of multi-source audio across 99 languages. However, less than 0.1% of its training corpus originated from sub-Saharan Africa, and zero hours came from Cameroonian languages. Because Whisper completely lacks language identification tokens and acoustic training representations for Ewondo, Basaa, or Ghɔmálá', applying off-the-shelf Whisper to these idioms results in transcription failure—the model either produces hallucinations in unrelated European languages or outputs silence.
- **Meta MMS (Massively Multilingual Speech) [7]:** Meta expanded pre-trained wav2vec 2.0 models across 1,400 languages using religious scripture readings from Faith Comes By Hearing (FCBH). For Cameroonian languages including Ewondo (`ewo`), Basaa (`bas`), and Ghɔmálá' (`bbj`), MMS included read-speech scripture audio. However, **zero open, conversational, or video-aligned colloquial speech corpora with standardized GACL orthography existed**. Because scripture read-speech differs sharply from informal video dialogue and the base MMS output vocabulary lacks GACL tone-marking diacritics, off-the-shelf MMS is not suited for conversational video captioning

Table 3: Systematic Literature Search Strategy and Record Retrieval Across Academic Databases

| Database | Search Query | Years | Records |
|---|---|---|---|
| IEEE Xplore | `("Automatic Speech Recognition" OR "ASR") AND ("low-resource" OR "tonal language") AND ("African languages" OR "Bantu")` | 2006–2026 | 64 |
| ACL Anthology | `("speech recognition" OR "speech corpus") AND ("African languages" OR "Cameroon" OR "Ewondo" OR "Basaa" OR "Ghomala")` | 2006–2026 | 78 |
| ACM Digital Library | `("video captioning" OR "subtitle editor" OR "Web Audio") AND ("multilingual" OR "waveform" OR "speech alignment")` | 2010–2026 | 49 |
| Scopus | `TITLE-ABS-KEY(("African languages" OR "tonal phonology") AND ("ASR" OR "corpus" OR "orthography" OR "diacritics"))` | 2006–2026 | 87 |
| Google Scholar | `"General Alphabet of Cameroonian Languages" OR "GACL diacritics" OR "KoboToolbox speech collection"` | 2006–2026 | 64 |
| **Total Records Retrieved Across All 5 Databases** | | | **342** |

Table 4: PRISMA 2020 Study Screening, Eligibility Review, and Selection Flow

| PRISMA Selection Step | Record Count | Methodological Action and Exclusion Justification |
|---|---|---|
| **1. Initial Identification** | $N = 342$ | Identified through structured Boolean search across the five academic databases. |
| **2. Deduplication** | $n = 258$ | Removed 84 identical cross-indexed duplicate records. |
| **3. Title and Abstract Screening** | $n = 92$ | Removed 166 records failing relevance criteria $EC_1$, $EC_2$, and $EC_4$. |
| **4. Full-Text Eligibility Review** | $n = 41$ | Excluded 51 articles: 24 lacked low-resource testing, 17 lacked reproducible data, and 10 lacked technical depth. |
| **5. Included Primary Studies** | **n = 41** | Final verified empirical corpus retained for systematic analysis and benchmarking. |

without downstream adaptation.

**Methodological Baseline and Hypotheses for Chapter 5.** Rather than assuming unverified baseline metrics prior to experimental testing, we formulate our performance expectations as testable research hypotheses to be empirically benchmarked in Chapter 5:

- **Hypothesis $H_1$ (ASR Baseline Error and CTC Adaptation):** We hypothesize that evaluating off-the-shelf MMS on conversational Cameroonian video dialogue will yield a baseline Word Error Rate exceeding 75%, with tone confusion and diacritic omission representing the dominant error categories. Downstream fine-tuning with character-level CTC using a dedicated GACL vocabulary is hypothesized to reduce Word Error Rate below 35% and Character Error Rate below 15%.

### 2.3.2 Architectural Causes of Model Failure on Cameroonian Speech

Our empirical investigation identifies three primary architectural causes for these failures:

**1. Lack of Pitch Prioritization in Latent Acoustic Encoders.** Cameroonian languages are fundamentally tonal. In non-tonal languages like English or French, pitch variation conveys pragmatic emphasis or interrogative mood. In contrast, in Cameroonian languages, relative vocal pitch (high, mid, low) carries lexical and grammatical meaning [19–21, 23]:

- In **Ewondo** [21], pitch contrasts distinguish lexical minimal pairs: *káb* (high tone) denotes "antelope", whereas *kàb* (low tone) denotes "to tear or rip". Similarly, *mɔ́* (high tone) denotes "hand", whereas *mɔ̀* (low tone) denotes "child". Pitch inflection on verbal prefixes also signals grammatical tense.
- In **Basaa** [23], pitch determines lexical semantics: *hɛ́* (high tone) signifies "knife", whereas *hɛ̀* (low tone) signifies "to surpass or excel".
- In **Ghɔmálá'** [19, 20], lexical minimal pairs rely entirely on tone register: *tsó* (high tone) signifies "to pierce", whereas *tsò* (low tone) signifies "to pay or refund".

Standard foundation encoders such as wav2vec 2.0 or Whisper process raw waveforms or log-mel filterbanks. However, because their self-supervised objectives were optimized on predominantly non-tonal corpora, their latent representations do not prioritize fundamental frequency ($F_0$) contours, and their downstream decoding vocabularies lack tone-marked tokens. Consequently, encoders map tone-differentiated minimal pairs to identical acoustic embeddings.

**2. Sub-Word Tokenizer Fragmentation of Combining Diacritics.** To convert continuous acoustic features into written text, neural decoders frequently employ sub-word tokenizers such as Byte-Pair Encoding (BPE). The 1979 General Alphabet of Cameroonian Languages (GACL) [5] introduces specialized vowels (ɔ, ə) that require stacked diacritics to represent pitch registers. Standard sub-word tokenizers treat combining diacritical marks as independent punctuation tokens, cleaving the accent away from its base vowel [25]. This creates fragmented, unreadable character sequences and severely inflates Character Error Rates (CER).

**3. The Unicode Normalization Dilemma (NFC vs. NFD).** Under the Unicode Standard [8], accented characters can be encoded in two distinct forms:

1. **Canonical Composition (NFC):** Represents character and diacritic as a single unified code point where defined.
2. **Canonical Decomposition (NFD):** Decomposes the character into a base glyph code point followed by one or more combining diacritical code points (e.g., `U+0254` + `U+0301`).

In Cameroonian orthographies, many stacked diacritic combinations lack precomposed NFC code points. If the acoustic vocabulary contains NFD sequences while reference transcripts use NFC, or vice versa, the ASR loss function penalizes correct transcriptions as deletion or substitution errors. Establishing strict Unicode normalization (enforcing NFC with deterministic combining mark ordering) is therefore a critical system design requirement for Chapter 3.

### 2.3.3 The Character-Level CTC Solution

To overcome tokenizer fragmentation without requiring prohibitive compute budgets, the most effective approach is fine-tuning a multilingual encoder (such as MMS) using **Connectionist Temporal Classification (CTC)** [3, 4]. CTC maps audio representations directly to discrete character tokens. By expanding the model's vocabulary to treat accented letters as atomic units, CTC outputs complete, correctly accented characters without splitting them apart.

## 2.4 The General Alphabet of Cameroonian Languages and Font Rendering

This section answers $\mathbf{RQ_{Rev}3}$ and $\mathbf{RQ_{Rev}7}$ by examining how local orthographies interact with font rendering engines and commercial video editors.

### 2.4.1 The 1979 GACL Unified Writing System

Cameroon has 281 national languages [18], each possessing distinct phonology and tonal registers. In 1979, Cameroonian linguists Maurice Tadadjeu and Etienne Sadembouo introduced the **General Alphabet of Cameroonian Languages (GACL)** [5] to establish a standardized orthography so that every national language could be written without ambiguity.

The GACL introduced dedicated phonetic letters including open-o (ɔ, indicating an open back vowel), schwa (ə, indicating a mid-central vowel), and the velar nasal (ŋ). To indicate pitch, diacritical marks are placed directly above vowels: acute accents (´) for high tone, grave accents (`) for low tone, and circumflex accents (ˆ) for falling tone.

### 2.4.2 Technical Audit of Commercial Video Editors and WebVTT Renderers

An architectural examination of standard video captioning and editing suites—including Adobe Premiere Pro (2024),

DaVinci Resolve 18, YouTube Studio Subtitle Editor, Subtitle Edit (v4.0), and Aegisub (v3.2)—reveals widespread typographic failure when handling African orthographies with stacked diacritics:

- **Glyph Collision and Vertical Bounding Box Clipping:** Standard subtitle rendering boxes allocate line-height for standard European ascenders and descenders. Stacked combining accents that extend above the baseline cap-height collide with base vowels, clip outside the top boundary of subtitle bounding boxes, or trigger substitution with empty rectangular "tofu" replacement glyphs.
- **Loss of Vertical Positioning in Standard Timed Text Formats:** When subtitle files are exported to WebVTT (`.vtt`) or SubRip (`.srt`), browser video players typically discard custom vertical positioning offsets, preventing stacked diacritics from displaying cleanly for native audiences [11].

In Chapter 5, we will formally test and quantify font rendering fidelity across video players. To resolve these failures structurally, KCS Media Studio embeds dedicated web font stacks (Charis SIL and Noto Sans) paired with dynamic CSS line-height calculations that guarantee vertical headroom for multi-tier stacked accents.

## 2.5 African NLP and Regional Village Dialect Continua

This section answers $\mathbf{RQ_{Rev}4}$ by investigating regional language variation and examining software mechanisms to accommodate village dialects.

Recent scholarship by African natural language processing researchers, particularly within the Masakhane community [14, 28], demonstrates that language tools must be built in active collaboration with native speakers. Community members must act as authoritative validators rather than passive data subjects.

### 2.5.1 The Reality of Village Dialect Continua

A core limitation in global multilingual AI benchmarks (such as FLORES-200 [29] or NLLB [30]) is the false assumption that an African language is spoken uniformly across all regions. In reality, Cameroonian languages exist along regional dialect continua:

- **Ewondo (Beti-Pahuin Group, Guthrie A72):** Marked lexical and phonetic differences separate the urban vehicular variety of Yaoundé from the Mbalmayo variety and rural peripheral dialects [21, 22].
- **Ghɔmálá' (Eastern Grassfields Bamileke):** Bandjoun (Jo) serves as the primary written reference, whereas Bafoussam (Fusap) represents a rapid commercial variety, and Baham (Hom) introduces notable vowel and pitch shifts [19, 20].
- **Basaa (Narrow Bantu, Guthrie A43):** Central Basaa (Nyangon) aligns with established dictionaries [23], while the Babimbi dialect along the Sanaga River preserves older prefix structures and unique idioms [22].

### 2.5.2 Decoupled Dialect Architecture in KCS Media Studio

When software enforces a single rigid dialect, speakers of regional variants find the tool culturally alien and resist adoption [22]. However, training separate speech models for every single village is impossible due to severe data scarcity. The optimal architectural solution is a **decoupled design**:

1. A centralized acoustic model is fine-tuned on the most stable reference dialect (Bandjoun for Ghɔmálá', Yaoundé for Ewondo, and Central for Basaa).
2. The user interface provides localized village glossaries and real-time dialect toggles, allowing editors to swap standardized terms for local village variants without retraining the acoustic model.

## 2.6 Field Data Acquisition, 9-Parameter Nomenclature, and Data Governance

This section answers $\mathbf{RQ_{Rev}5}$ and $\mathbf{RQ_{Rev}6}$ by describing structured field recording protocols, file naming conventions, and double-validation quality control.

Prior to this project, zero hours of publicly available, time-aligned conversational speech data existed for Ghɔmálá', Ewondo, or Basaa in standardized GACL orthography. While comprehensive grammar books and dictionaries exist [5, 21, 23], print references alone cannot train speech recognition models.

### 2.6.1 Offline Mobile Collection with KoboToolbox and Mobile Tooling

Fieldwork in rural Cameroon takes place in communities with unstable electricity and absent cellular data coverage. Cloud-dependent recording platforms fail in these settings. Following best practices established by mobile language documentation initiatives like Aikuma [31] and radio-based corpus collection [32], we utilize **KoboToolbox** [33], an open-source offline survey platform maintained by the Harvard Humanitarian Initiative.

Field enumerators use Android smartphones running custom KoboToolbox forms. The form collects participant demographic metadata, records verbal and signed informed consent [13, 34], presents prompts from our 363-word diagnostic lexicon, and records audio locally. When devices re-enter areas with network connectivity, recordings automatically synchronize with the central project repository.

### 2.6.2 The 9-Parameter File Nomenclature and Speaker Leakage Prevention

Arbitrary file naming causes metadata loss and file overwrites. To guarantee full machine-actionable indexing, every audio file follows the strict **nine-parameter naming convention** established in Assignment 1:

```
ISO-Lang_Variant_Demographics_Content_-
Environment_Location_Timestamp_ConceptID_-
UID.wav
```

For example:

```
ewo_yao_ADL-M_LEX_QUI_YAO_20260911_-
ASD-0001_1a2b3c4d.wav
```

This file name encodes language (Ewondo), dialect (Yaoundé), speaker demographics (Adult Male), content type (Lexical prompt), acoustic environment (Quiet room), geographic location (Yaoundé), timestamp (`YYYYMMDD`), concept identifier, and a unique cryptographic hash.

**Governing Train-Test Splits to Prevent Speaker Leakage.** A critical methodological vulnerability in machine learning speech research is *speaker leakage*—where recordings from the same individual appear in both the training set and the evaluation set, producing falsely optimistic accuracy scores [35]. Although our 9-parameter filename contains an anonymized demographic code and unique file hash, file names alone cannot enforce speaker independence.

To guarantee complete partition integrity, speaker identity is managed through an **external relational Speaker Registry table** linked to documented participant consent forms. Each participant is assigned a unique, anonymized speaker identifier (`speaker_id`). Partitioning algorithms allocate all recordings belonging to a given `speaker_id` strictly to either the training, validation, or test split, eliminating speaker leakage completely.

### 2.6.3 Informed Consent and Double-Validation Workflow

Ethical fieldwork requires documented informed consent prior to recording. Contributors are informed of data usage for open research and software development, and retain the right to withdraw recordings at any time [13, 34].

To prevent corrupted audio or erroneous spellings from entering the model training set, all field recordings pass through an **independent two-tier validation loop**:

1. **Tier 1 (Technical Quality Audit):** A technician verifies that the recording is free of distortion, clipping, and ambient noise.
2. **Tier 2 (Linguistic Verification):** An independent native speaker verifies that the pronunciation is authentic and matches official GACL dictionary orthography.
3. **Commitment:** Only audio passing both verification tiers is admitted into the official training and evaluation corpora.

## 2.7 Video Subtitle Editors, Bantu Text Expansion, and Browser Audio Waveforms

This section answers $\mathbf{RQ_{Rev}7}$ and $\mathbf{RQ_{Rev}8}$ by explaining how timeline desynchronization arises from Bantu text expansion and how browser waveforms solve it.

### 2.7.1 Timeline Desynchronization from Bantu Text Expansion

Subtitles make audiovisual content accessible to students, hard-of-hearing audiences, and multilingual viewers [15]. However, translating European text into Bantu languages like Ewondo or Basaa triggers **text expansion of 30% to 50%** [21, 23]. Bantu morphology relies on agglutinative noun-class concord prefixes and verbal extensions, naturally producing longer phrases. In standard subtitle tools, translated subtitle blocks expand past the speaker's vocal

duration, spilling across scene boundaries and falling out of sync with video dialogue [16, 17].

### 2.7.2 The Manual Subtitling Productivity Bottleneck

Because existing editing suites offer zero automated transcription or dialect-specific input support for Cameroonian languages, video subtitling is currently performed entirely by hand. Translators must repeatedly switch between video players and word processors, search external character tables to copy-paste stacked diacritics, and manually align subtitle boundaries against video frames. Professional audiovisual subtitling guidelines and preliminary translator interviews indicate that manual transcription, translation, and block-by-block synchronization for unassisted low-resource speech typically imposes a heavy overhead of approximately 10 to 12 minutes of labor per minute of finished video dialogue. The vast majority of this overhead is absorbed by character map lookups, correcting clipped font renderings, and manually nudging misaligned caption blocks.

### 2.7.3 Browser Waveforms and Automated Boundary Snapping

To eliminate this bottleneck, the KCS Media Studio combines the browser-native **Web Audio API** [10] with an **HTML5 Canvas** timeline. By decoding audio buffers client-side, the Web Audio API targets sub-15ms interactive rendering latencies (meeting W3C specifications for imperceptible audio scrubbing) without relying on high-latency round-trip server requests.

When the speech model transcribes an utterance, forced alignment techniques [4, 36] identify exact voice onsets and pauses. Subtitle blocks automatically snap to these speech boundaries on the waveform, giving translators immediate visual feedback and cutting editing overhead dramatically.

## 2.8 Measuring System Success and Usability

This section answers $\mathbf{RQ_{Rev}9}$ by establishing the objective accuracy scores and usability benchmarks used to evaluate the KCS Media Studio:

### 2.8.1 Objective Speech Recognition Scores

- **Word Error Rate (WER):** The standard metric measuring substituted, deleted, and inserted words against human reference transcripts.
- **Character Error Rate (CER):** Measures letter-by-letter transcription accuracy. For tonal languages, CER is the most telling metric: because changing a single accent mark changes word meaning completely, we must evaluate letter and accent accuracy independently of whole-word boundaries.

### 2.8.2 Productivity and Usability Metrics

- **Subtitling Throughput (min/min video):** The working time required to produce a verified subtitle track per minute of video dialogue. While unassisted manual subtitling in indigenous languages often requires 10 or more minutes of manual labor per minute of video, our target hypothesis for KCS Media Studio is to achieve $<$**3.5 min/min video** (a projected productivity improvement exceeding 65%). This efficiency metric will be empirically validated in Chapter 5 through timed subtitling trials with native translators.
- **System Usability Scale (SUS) [37]:** A standardized 10-item psychometric survey of overall system usability. In Chapter 5, we target an empirical score above **80/100** across native translator cohorts.
- **NASA Task Load Index (NASA-TLX) [38]:** Measures subjective workload across six cognitive dimensions (Mental Demand, Physical Demand, Temporal Demand, Performance, Effort, Frustration), verifying that localized dialect toggles reduce cognitive burden.

## 2.9 Concrete Systems and Tooling Comparison Matrix

To address the requirement for a rigorous, shared-criteria comparison of concrete systems, Table 5 evaluates eight prominent speech recognition models, corpora, and subtitle editing suites across eight standardized dimensions.

## 2.10 Systematic Comparative Synthesis of Existing Solutions

To synthesize findings across all nine review dimensions, Table 6 presents a direct comparison between current state-of-the-art technologies and the unique requirements of indigenous Cameroonian languages.

Across all areas—speech recognition, font rendering, dialect handling, rural audio collection, and subtitle synchronization—existing commercial suites demonstrate

Table 5: Concrete System Comparison Matrix: Speech Models, Corpora, and Subtitling Suites Evaluated on Shared Criteria

| System / Suite | Architecture / Data Foundation | African / Cameroonian Coverage | Tonal Pitch Modeling ($F_0$) | GACL Diacritic Rendering | Dialect Support | Evaluation Metrics | Primary Limitation for KCS Project |
|---|---|---|---|---|---|---|---|
| **OpenAI Whisper (Large-v3)** [2] | Transformer Seq2Seq (680k hrs multi-lingual) | 99 languages; 0 Cameroonian languages | Discarded in encoder representations | Unsupported; tokenization splits combining accents | None | WER / BLEU | Complete transcription failure; produces hallucination or silence due to absent language tokens. |
| **Meta MMS (1.4k+)** [7] | wav2vec 2.0 + CTC (FCBH New Testament) | 1,400+ languages; includes `ewo`, `bas`, `bbj` read speech | Partial acoustic representation; lacks tone labels | Splitting under default subword tokenizer | Single standard per language | Character-level / Word-level WER | Trained on scripture read-speech; lacks tone labels in output vocabulary; unadapted to conversational speech; no web UI. |
| **Meta XLS-R (53/128)** [6] | Self-supervised wav2vec 2.0 (436k hrs) | 128 languages; limited sub-Saharan data | Acoustic latent space captures formants, not pitch | Dependent on downstream tokenizer | None | Cross-lingual WER transfer | Requires extensive supervised fine-tuning; no integrated subtitle editor or UI. |
| **AfriSpeech-200 / FLEURS** [39, 40] | Curated read speech benchmarks | 13+ African languages; no Central African tonal sets | Acoustic evaluation only | Standard Latin without stacked accents | Regional accent tags (non-Cameroonian) | WER / CER / PER | Acoustic evaluation benchmark only; provides no video editing or captioning software. |
| **Subtitle Edit (v4.0)** | Desktop C# WinForms open-source editor | Language-agnostic string editing | None (manual timing) | Partial; diacritic collisions on default fonts | None | Manual timing accuracy | Desktop-only (not web-based); lacks integrated Cameroonian ASR and dialect toggles. |
| **Aegisub (v3.2)** | Desktop C++ ASS subtitle suite | Language-agnostic string editing | None (audio waveform display) | Frequent vertical clipping in bounding boxes | None | Visual frame alignment | Requires complex manual installation; zero ASR integration; desktop legacy code. |
| **YouTube Studio Subtitle Editor** | Proprietary cloud video platform | 100+ languages; 0 Cameroonian languages | Cloud ASR (non-tonal) | WebVTT export discards vertical offsets | None | Subtitle completion time | Commercial closed-source; zero support for GACL orthographies or local dialects. |
| **ELAN (Max Planck)** [41] | Professional linguistic tier annotation tool | Universal Unicode tier annotation | External plugin required (Praat) | Full Unicode support (requires SIL fonts) | Multi-tier manual dialect notes | Tier alignment precision | High cognitive barrier for video editors; zero automated ASR caption snapping. |

critical limitations when applied to Central African languages. These limitations are fundamental architectural barriers rather than superficial styling gaps. Table 6 summarizes these shortcomings and defines the corresponding architectural choices for KCS Media Studio.

## 2.11 Evidence-Based Research Gap and 10 Practical System Challenges

### 2.11.1 The Evidence-Based Research Gap

The reviewed literature confirms substantial progress in cross-lingual speech representation transfer (XLS-R, MMS), participatory African NLP corpus creation (Masakhane), and client-side browser digital signal processing. However, the evidence reveals a critical scientific and technical research gap: **there exists no empirically evaluated, consent-governed transcription and captioning workflow for Cameroonian tonal languages (Ghɔmálá', Ewondo, Basaa) that simultaneously resolves Unicode GACL diacritic stacking, accommodates regional dialect continua, and prevents timeline desynchronization under Bantu morphological text expansion.**

### 2.11.2 10 Practical System Challenges for Chapter 3

To bridge this overarching research gap, we translate our empirical findings into ten concrete engineering challenges ($C_1$–$C_{10}$) that directly drive the system architecture in Chapter 3:

Table 6: Systematic Comparative Synthesis: State-of-the-Art Capabilities, Limitations, and KCS Media Studio Architectural Choices

| Research Area | Current State of the Art | Critical Limitation for KCS Project | KCS Media Studio Architectural Choice |
|---|---|---|---|
| **Speech AI ($\mathbf{RQ_{Rev}}1$, $\mathbf{RQ_{Rev}}2$)** | Self-supervised models (wav2vec 2.0, MMS, Whisper) pre-trained across hundreds of languages [2, 7, 24]. | Severe baseline error degradation; models lack tonal pitch discrimination; sub-word tokenizers fragment diacritics. | Fine-tune MMS using character-level CTC with a dedicated GACL character vocabulary so accents remain bound to base vowels. |
| **Orthography & Fonts ($\mathbf{RQ_{Rev}}3$)** | Official GACL standard established in 1979 [5]; Unicode combining diacritic character definitions [8]. | Stacked accents collide with letters, clip outside subtitle boxes, or render as "tofu" glyphs in standard editors. | Embed dedicated UTF-8 font stacks (Charis SIL, Noto Sans) with adaptive CSS line-height bounding boxes. |
| **African NLP & Dialects ($\mathbf{RQ_{Rev}}4$)** | Community benchmarks (FLORES-200, Masakhane) with participatory native speaker validation [14, 28, 29]. | Benchmarks assume a single uniform dialect; tools lack dialect continuum support and editor integration. | Decouple acoustic decoding from regional vocabularies; implement localized village glossaries and UI dialect toggles. |
| **Fieldwork & Audio Quality ($\mathbf{RQ_{Rev}}5$, $\mathbf{RQ_{Rev}}6$)** | Offline mobile survey forms via KoboToolbox; mobile audio acquisition [31, 33]. | Unstructured field audio risks missing metadata, naming collisions, and train-test data leakage. | Enforce a strict 9-parameter file nomenclature and relational speaker ID table with two-tier double-validation. |
| **Video Subtitle Editors ($\mathbf{RQ_{Rev}}7$, $\mathbf{RQ_{Rev}}8$)** | WebVTT track editing; standard browser video playback [11, 16, 17]. | GACL diacritics get clipped; translation causes 30%–50% text expansion; unassisted manual subtitling requires $\sim$10 min/min video. | Build an in-browser Web Audio API and HTML5 Canvas waveform editor with automated speech boundary snapping. |
| **Testing & Usability ($\mathbf{RQ_{Rev}}9$)** | Objective NLP metrics (WER, CER) and human usability surveys (SUS, NASA-TLX) [37, 38]. | Whole-word error metrics overlook accent mistakes; usability benchmarks rarely evaluate subtitling throughput speed. | Measure letter-level CER, track subtitling throughput ($<$3.5 min/min), and verify translator approval with native cohorts. |

1. **Challenge $C_1$ (Tonal Pitch Preservation in Acoustic Modeling):** In Cameroonian languages, vocal pitch carries lexical and grammatical meaning. For example, in Ewondo, *káb* (high tone, "antelope") contrasts with *kàb* (low tone, "to tear"); in Basaa, *hɛ́* ("knife") contrasts with *hɛ̀* ("to surpass"); and in Ghɔmálá', *tsó* ("to pierce") contrasts with *tsò* ("to refund") [20, 21, 23]. Because global foundation models were pre-trained predominantly on non-tonal European languages, their latent acoustic representations do not prioritize pitch contours ($F_0$). In Chapter 3, we resolve this by fine-tuning Meta MMS using Connectionist Temporal Classification (CTC) with an explicit tone-differentiated character inventory.

2. **Challenge $C_2$ (Preserving Stacked Diacritics via Unicode NFC Normalization and Atomic CTC):** The 1979 GACL orthography requires accent marks stacked on special vowels (ɔ, ə). Standard sub-word tokenizers cleave combining diacritics away from base characters, inflating Character Error Rates. Furthermore, discrepancies between Canonical Composition (NFC) and Canonical Decomposition (NFD) cause character mismatch during training. Chapter 3 addresses this by enforcing strict NFC normalization and configuring an atomic character dictionary that keeps accented glyphs intact during acoustic decoding.

3. **Challenge $C_3$ (Diacritic-Safe Typography and Glyph Rendering):** When displaying GACL text in standard video editors, stacked accents collide with letters, clip outside subtitle bounding boxes, or render as replacement tofu boxes due to absent Central African glyph tables. In Chapter 3, we embed a dedicated web font stack (Charis SIL and Noto Sans) with adaptive CSS line-height calculations that guarantee zero clipping across all subtitle tracks.

4. **Challenge $C_4$ (Dialect Continuum Accommodation via Decoupled Glossaries):** Spoken idioms vary naturally across village continua: Ewondo differs between Yaoundé and Mbalmayo; Ghɔmálá' varies between Bandjoun, Bafoussam, and Baham; and Basaa differs between Central and Babimbi. Forcing a single rigid orthography alienates native speakers, yet training separate neural models for every village is impossible due to scarce training data. In Chapter 3, we solve this with a decoupled architecture: a shared acoustic model transcribes speech sounds, while the user interface provides localized village glossaries and real-time dialect toggles.

5. **Challenge $C_5$ (Offline Rural Audio Acquisition and Network Resilience):** Gathering high-quality speech data requires recording native speakers in rural villages lacking cellular internet and stable electricity. Standard web recording apps fail when disconnected. In Chapter 3, we deploy an offline mobile collection pipeline using KoboToolbox on field smartphones, storing audio securely on device and automatically synchronizing when internet connectivity is restored.

6. **Challenge $C_6$ (9-Parameter Nomenclature and Relational Speaker Split Governance):** Arbitrary file naming leads to metadata loss and data corruption. Furthermore, preventing train-test data leakage requires partitioning recordings at the speaker level rather than the file level. In Chapter 3, we implement an automated ingestion service enforcing our strict 9-parameter naming convention linked to an external relational Speaker Registry table.

7. **Challenge $C_7$ (Two-Tier Technical and Linguistic Double-Validation):** Field recordings can suffer from acoustic distortion or pronunciation inconsistencies. Because automated scraping cannot verify linguistic authenticity, Chapter 3 establishes a mandatory two-tier quality control workflow: every recording is audited first by a technician for acoustic clarity and second by an independent native speaker for authentic pronunciation and GACL spelling before entering the training corpus.

8. **Challenge $C_8$ (Bantu Text Expansion and Timeline Desynchronization):** Translating concise European dialogue into Bantu languages expands text length by 30% to 50% due to agglutinative noun-class agreements and verbal extensions [21, 23]. In video subtitles, these expanded sentences spill across scene cuts and fall out of sync with the speaker's voice. In Chapter 3, we implement automated subtitle re-segmentation and real-time visual expansion indicators to keep translated captions neatly aligned with speech.

9. **Challenge $C_9$ (Low-Latency Browser Waveforms and Audio Snapping):** Manually aligning subtitles by dragging text blocks on a timeline is slow and error-prone, requiring an estimated 10 to 12 minutes per minute of video in unassisted manual workflows. In Chapter 3, we utilize the browser Web Audio API and HTML5 Canvas to render interactive waveforms with sub-15ms latency, paired with automated forced alignment that snaps caption boundaries directly to voice onsets.

10. **Challenge $C_{10}$ (Translator Productivity and Cognitive Ergonomics):** Manual subtitling in indigenous languages currently forces translators to copy-paste special letters from external tables, causing cognitive fatigue. In Chapter 3, we design a clean, responsive web interface targeting a reduction in subtitling effort from an unassisted baseline of approximately 10 minutes down to under 3.5 minutes per minute of video, to be empirically validated through standardized System Usability Scale (target hypothesis: SUS $> 80$) benchmarks in Chapter 5.

### 2.11.3 Traceability Matrix: Connecting Chapter 1 to Chapter 3

To guarantee methodological rigor, Table 7 provides an explicit traceability matrix linking Chapter 1's foundational

goals, the empirical observations from Chapter 2, the ten practical challenges ($C_1$–$C_{10}$), and the resulting architectural requirements for Chapter 3. In accordance with the Nunamaker SDIS framework, Chapter 2 reports strictly Observation-phase objectives, while Theory Building and Implementation specifications are defined as forward requirements to be addressed in Chapter 3.

## 2.12 Chapter Summary

This chapter carried out the Observation phase of our research by examining 41 primary peer-reviewed studies and benchmarking existing speech recognition and video editing tools on Cameroonian languages. The investigation answered all nine review questions and identified the key reasons why existing systems fail: they filter out voice pitch, fragment stacked accent marks, ignore regional village dialects, lack offline field collection tools, and require exhausting manual alignment effort.

By translating these empirical findings into **ten practical system challenges ($C_1$–$C_{10}$)**, we have established a solid foundation for the Kingdom Communication Solutions (KCS) Media Studio. Chapter 3 will convert these ten challenges into a concrete, component-level system design: presenting our modular microservice architecture, database entity schemas, and responsive user interface workflows.

Table 7: Methodological Traceability Matrix: Connecting Chapter 1 Goals to Chapter 2 Observations and Chapter 3 Design Specifications

| Goal | Problem | Chapter 2 Scope | Chapter 2 Observation | Challenge | Target Chapter 3 Design Specification |
|---|---|---|---|---|---|
| **RQ1** | **PS2** | Observation (**RO1.1.O**) | Spoken varieties differ by village; forcing a single dialect alienates speakers [20, 22]. | **Challenge** $C_4$ (Dialect Continua) | Village Glossary Microservice and UI Dialect Switch Toggle (Addresses **RO1.2.T**, **RO1.3.I**). |
| | | Observation (**RO1.1.O**) | Translators waste substantial effort searching external tables for special characters [5]. | **Challenge** $C_3$ **&** $C_{10}$ (Typography & Speed) | Diacritic-safe Unicode Font Stacks and Ergonomic Caption Interface (Addresses **RO1.3.I**). |
| | | Observation (**RO1.1.O**) | Unstructured field audio causes demographic skew and train-test data leakage [35]. | **Challenge** $C_5$ **&** $C_6$ (Field Recording & Governance) | Offline KoboToolbox Ingestion API, 9-Parameter Rule, and Speaker Registry (Addresses **RO2.2.T**). |
| | | Observation (**RO1.1.O**) | Independent linguistic verification prevents dialect errors from corrupting training data. | **Challenge** $C_7$ (Quality Validation) | Two-Tier Validation State Machine and Consent Tracking Ledger (Addresses **RO2.2.T**). |
| **RQ2** | **PS1** | Observation (**RO2.1.O**) | Baseline foundation models lack GACL diacritics and pitch features; sub-words split accents [2]. | **Challenge** $C_1$ **&** $C_2$ (Pitch & Tokenization) | Character-level CTC Fine-Tuning with GACL Atomic Vocabulary and NFC (Addresses **RO2.3.I**). |
| | | Observation (**RO2.1.O**) | Translating into Bantu languages causes 30%–50% text expansion, spilling across video scene cuts [21]. | **Challenge** $C_8$ (Text Expansion) | Automated Subtitle Re-segmentation and Expansion Alert Indicators (Addresses **RO2.3.I**). |
| | | Observation (**RO2.1.O**) | Traditional server-side audio processing introduces lag; unassisted manual alignment takes $\sim$10 min/min video [10]. | **Challenge** $C_9$ (Waveform Visualization) | Browser Web Audio API Waveform Engine with Automated Boundary Snapping (Addresses **RO2.3.I**). |
| | | Observation (**RO2.1.O**) | Translator fatigue causes high error rates; usability scores must exceed global averages [37, 38]. | **Challenge** $C_{10}$ (Usability & Throughput) | Accelerated Subtitling Pipeline ($<$3.5 min/min video, SUS $>$ 80) (Addresses **RO2.4.E**). |

# Appendix: Systematic Literature Review Appraisal Table

Table 8 presents the quality appraisal of representative primary studies retained under our PRISMA methodology, documenting empirical evidence types, corresponding review questions, and quality ratings.

Table 8: Systematic Literature Review Appraisal: Representative Included Studies

| **Citation** | **Evidence Focus** | **Research Methodology** | **Review Question** | **Quality Appraisal / Empirical Rating** |
|---|---|---|---|---|
| Pratap et al. (2024) [7] | Massively Multilingual Speech (MMS 1,400+) | Neural self-supervised acoustic modeling | $RQ_{Rev}1$ | **High**: Large-scale empirical validation; verified checkpoint availability. |
| Radford et al. (2023) [2] | Robust Speech Recognition via Large-Scale Weak Supervision | Sequence-to-sequence Transformer evaluation | $RQ_{Rev}1$ | **High**: Authoritative 680k-hour benchmark; documents non-coverage of African idioms. |
| Olatunji et al. (2023) [39] | AfriSpeech-200 Accented African Speech Corpus | Multi-accent clinical speech recognition | $RQ_{Rev}1$, $RQ_{Rev}5$ | **High**: Verifiable Pan-African dataset; peer-reviewed evaluation protocols. |
| Conneau et al. (2023) [40] | FLEURS Multilingual Speech Benchmark | Few-shot cross-lingual representation testing | $RQ_{Rev}1$ | **High**: Rigorous 102-language baseline for self-supervised evaluation. |
| Gauthier et al. (2016) [35] | Low-Resource Speech Corpus Collection | Empirical field recording in Wolof | $RQ_{Rev}5$, $RQ_{Rev}6$ | **High**: Documents speaker leakage risks and field validation workflows. |
| Bird et al. (2014) [31] | Aikuma Mobile Language Documentation App | Field mobile recording in rural regions | $RQ_{Rev}5$ | **High**: Validated offline mobile acquisition architecture and usability. |
| Wittenburg et al. (2006) [41] | ELAN Multimodal Linguistic Annotation | Professional tier annotation architecture | $RQ_{Rev}7$ | **High**: Industry standard for time-aligned linguistic audio transcription. |
| Doumbouya et al. (2021) [32] | Radio Broadcasts for Low-Resource ASR | Field acquisition and crowdsourcing | $RQ_{Rev}5$ | **High**: Peer-reviewed AAAI study on acoustic collection in West Africa. |
| Gebru et al. (2021) [34] | Datasheets for Datasets | Dataset governance and ethical audits | $RQ_{Rev}6$ | **High**: Standardized framework for corpus transparency and consent documentation. |
| Tadadjeu & Sadembouo (1979) [5] | General Alphabet of Cameroonian Languages | National standard phonetic orthography | $RQ_{Rev}3$ | **High**: Foundational national standard for all Cameroonian language writing. |
| Essono (2000) [21] | Phonology, Morphology, and Syntax of Ewondo | Linguistic grammatical treatise | $RQ_{Rev}2$, $RQ_{Rev}4$ | **High**: Authoritative linguistic reference grammar for the Ewondo language. |
| Lemb & de Gastines (1973) [23] | Dictionnaire Basaa-Français | Lexicographical reference grammar | $RQ_{Rev}2$, $RQ_{Rev}4$ | **High**: Standard reference dictionary for Basaa tonology and vocabulary. |
| Sadembouo (1984) [20] | Dialectology and Communication in Bamileke | Sociolinguistic dialect survey | $RQ_{Rev}2$, $RQ_{Rev}4$ | **High**: Authoritative classification of Ghɔmálá' regional varieties. |

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